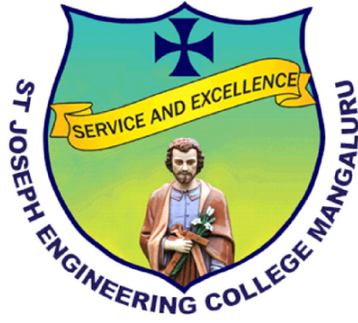




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Multimodal Agentic Framework for Behavioral Profiling Using Video and Audio Modalities

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Contents

- Introduction
- Purpose
- Objectives
- Methodology
- Results and Discussion
- References



Introduction

- Behavioral profiling is crucial in fields like psychology, security, and human-computer interaction for understanding emotions, intentions, and actions.
- Traditional methods often rely on limited data sources, reducing accuracy and effectiveness.
- Multimodal data—video (facial expressions, gestures) and audio (tone, pitch) provides a more comprehensive view of human behavior.
- Current profiling systems struggle to combine different data types, work in real time, and adjust to live interactions, which affects their accuracy and usefulness in real-world situations.



Purpose

To improve the efficiency of multimodal systems by reducing the frequency of video and audio calls to large language models, thereby ensuring cost-effective infrastructure management and faster response times in real-time multimodal use cases



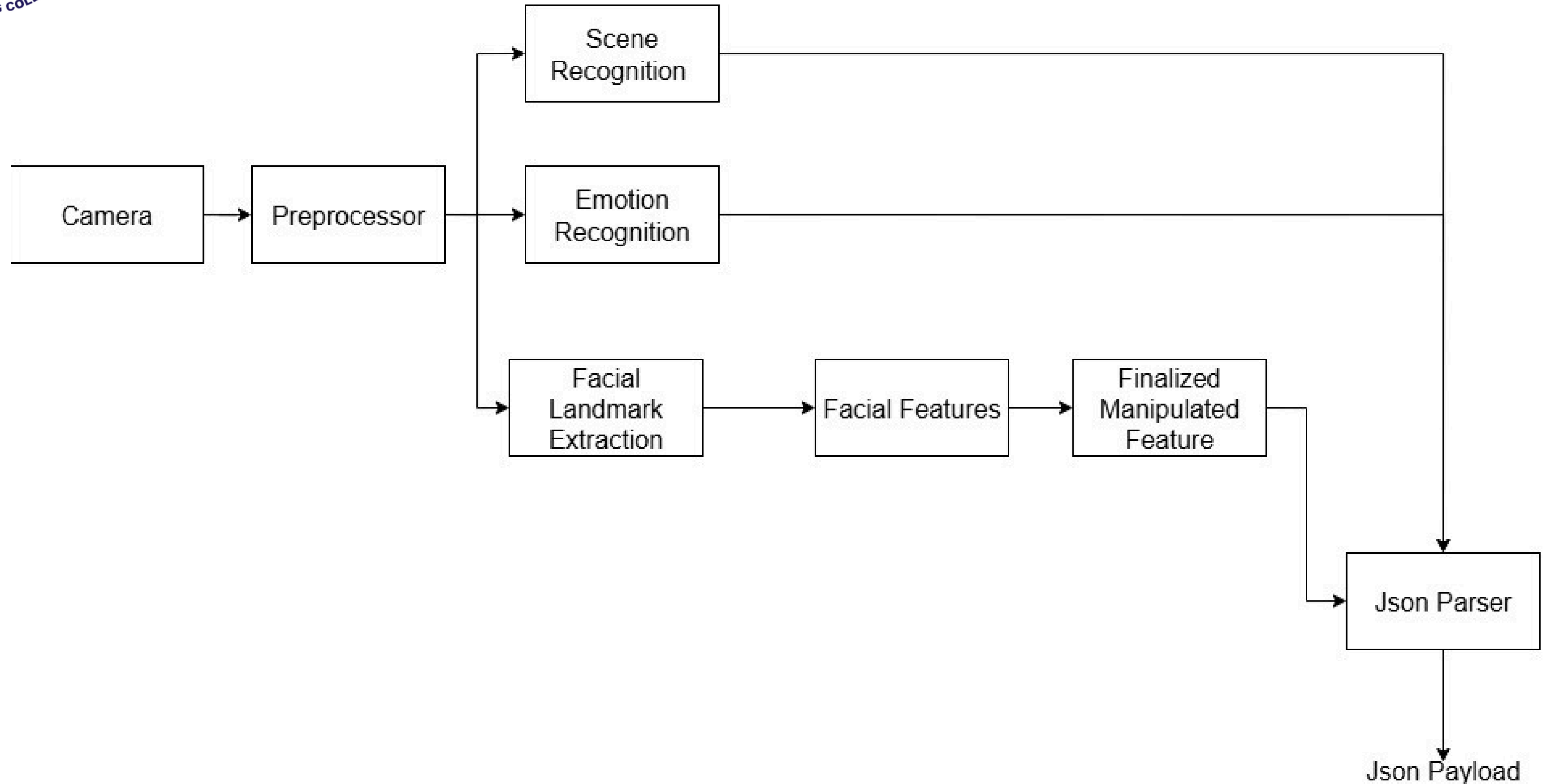


Objectives

1. To design a Camera Framework for capturing visual behavioral features.
2. To develop a audio processing model for spatial audio capture.
3. To establish an Agentic Orchestration Layer
4. To implement a Multimodal Framework fusing audio and video data.

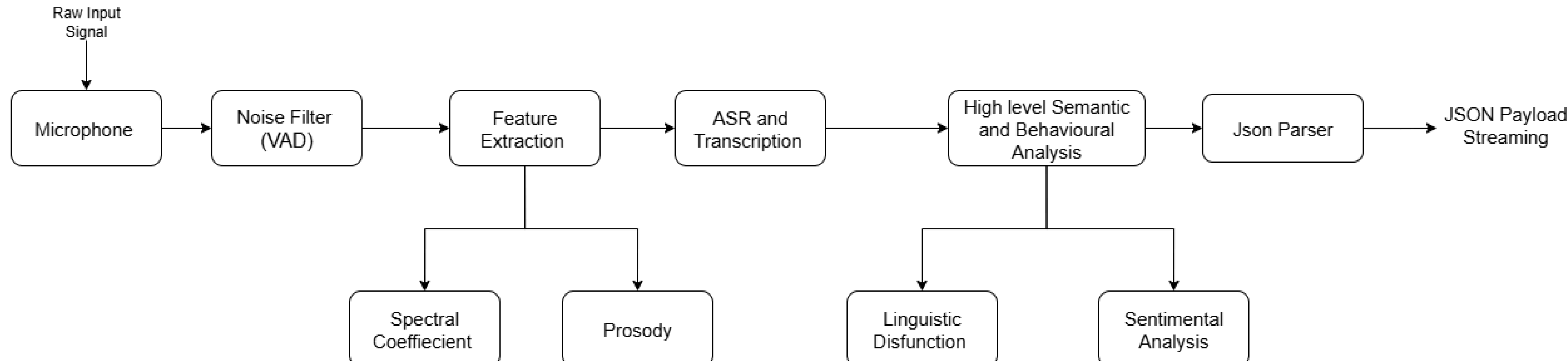


Video modality flowchart



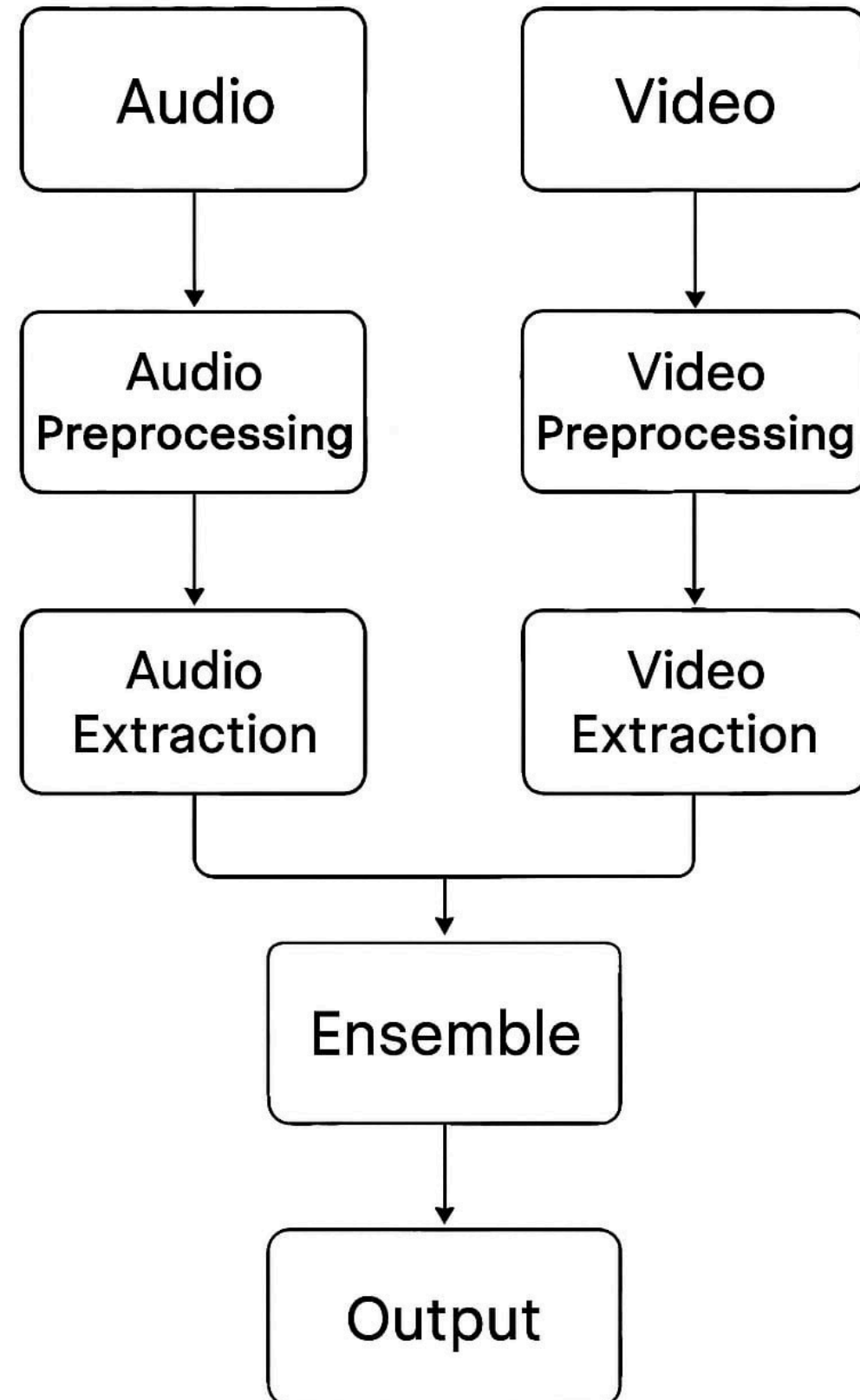


Audio Modality Flowchart



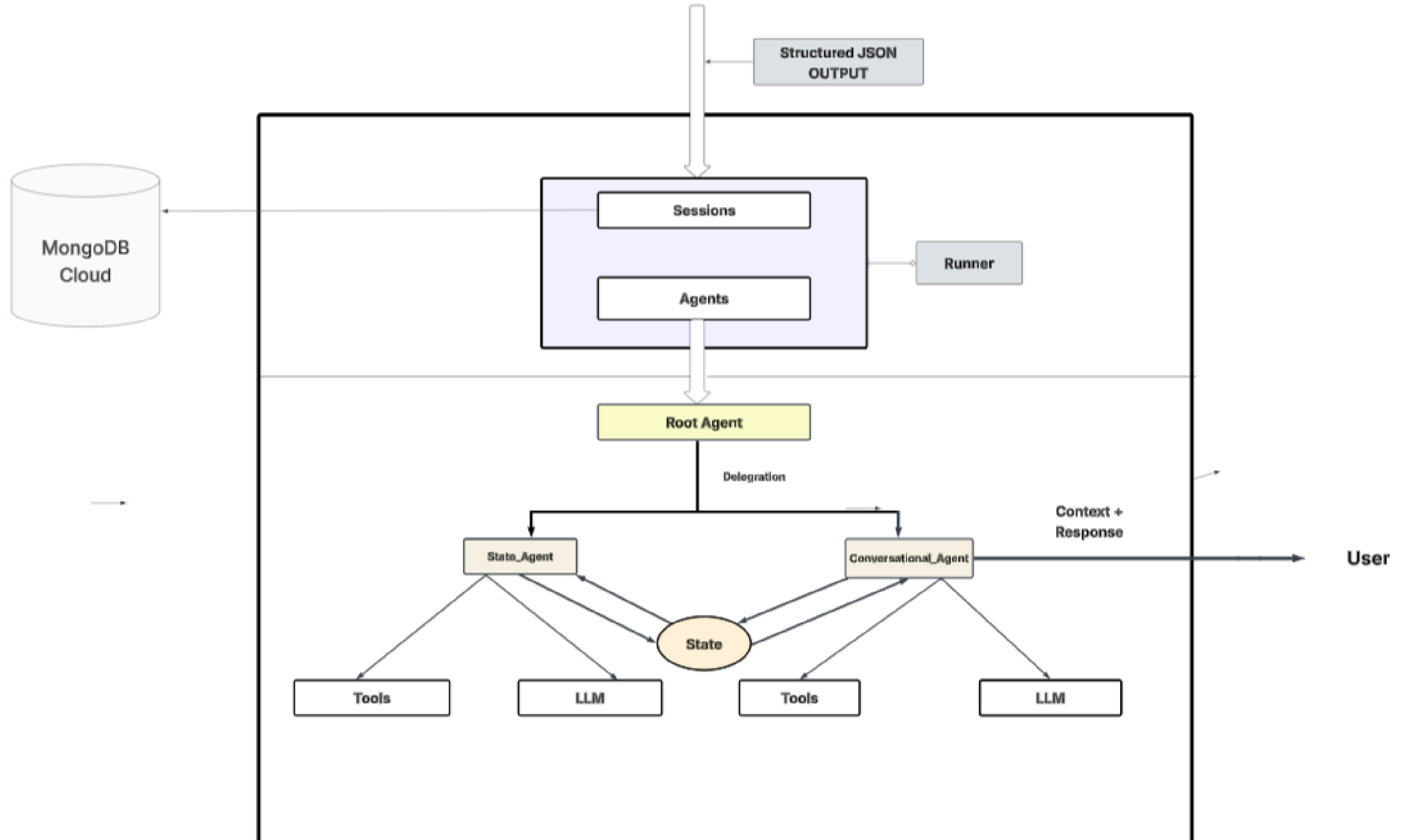


Model Pipeline



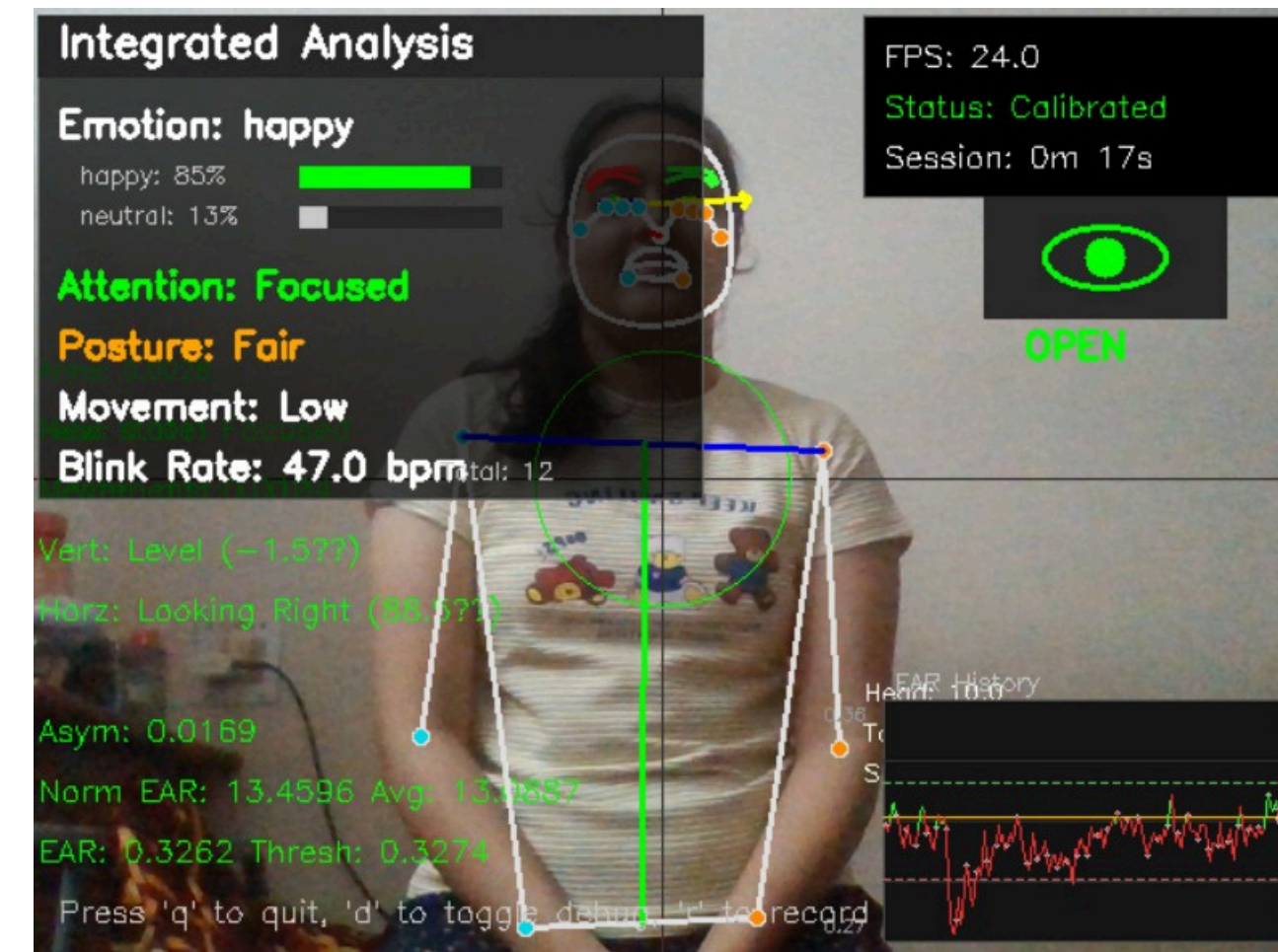
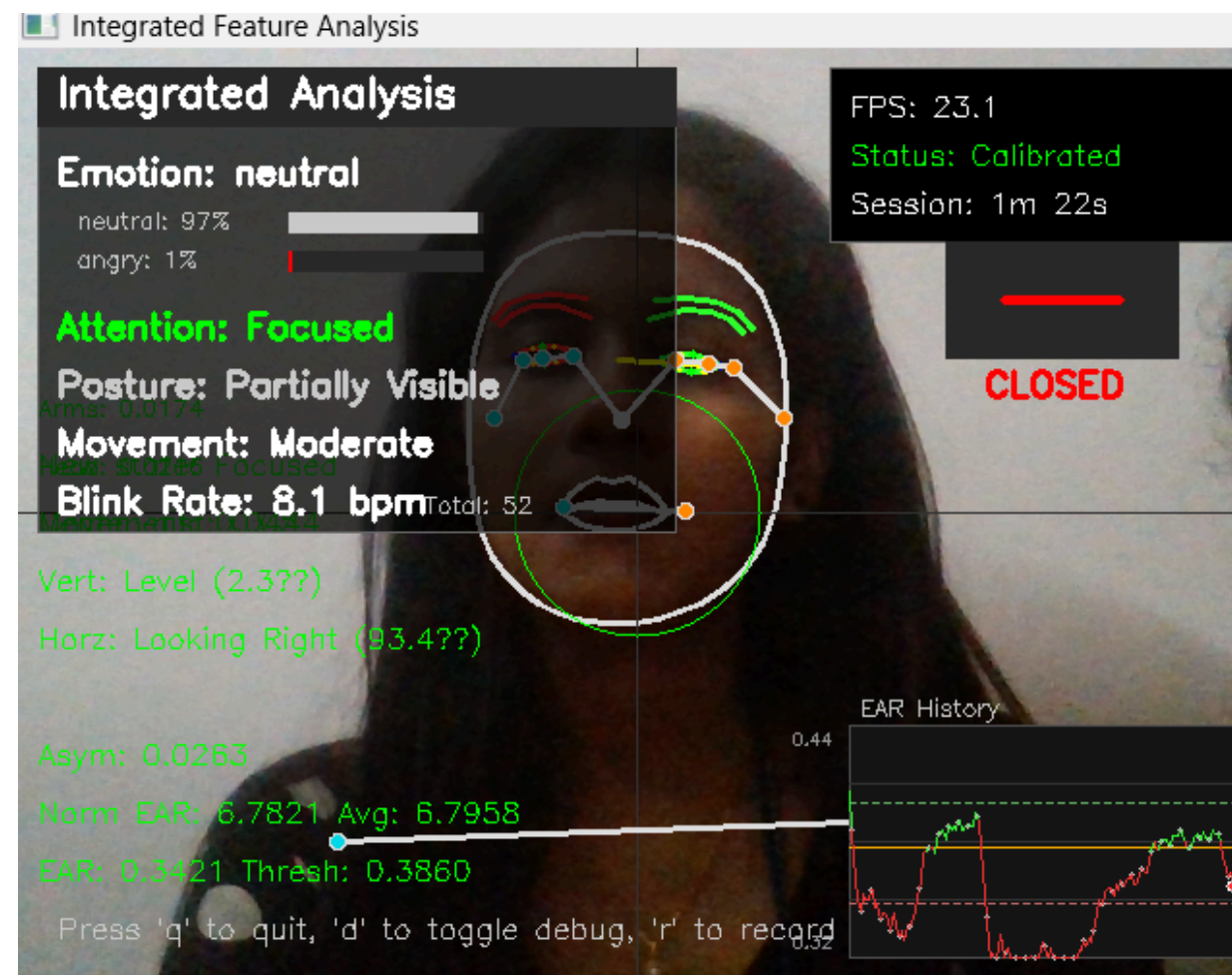


ADK





Result and Discussion



Result and Discussion

Realtime Multimodal Context

Connected to ws://127.0.0.1:8087/ws/context

Engagement	Speaking Prob	Valence	Arousal	Audio Rel	Video Rel
0.69	0.6	0.0	1.0	0.57	0.75

> Raw context

Events

—

Features (/ws/features)

Audio

ASR: —

ASR Conf

—

Pitch (Hz)

54.48

Energy (dB)

-56.03

Video

Face conf

0.8

Gaze on camera: True

Head [pitch, yaw]: [0.08333333333333333, -0.046875]

Blinking: False

Blink rate (bpm)

5.7

Scene: —

Realtime Multimodal Context

Connected to ws://127.0.0.1:8087/ws/context

Engagement	Speaking Prob	Valence	Arousal	Audio Rel	Video Rel
0.79	0.6	0.0	0.96	0.57	0.74

Raw context

```
{
  "ts": 1754707788.2543092,
  "session_id": "local",
  "engagement": 0.7913256784446738,
  "affect": {
    "valence": 0.0,
    "arousal": 0.9597623635915837
  },
  "speaking_prob": 0.5992026255347189,
  "events": [],
  "reliability": {
    "audio": 0.574610826185568,
    "video": 0.7446969696969694
  }
}
```

Events

—

Features (/ws/features)



Results and Discussion

- **High-Fidelity Visual Analysis:** Our camera framework captures complex visual behaviors in real-time (24 FPS), accurately detecting facial expressions ("happy"), attention ("Focused"), body posture ("Fair"), and eye metrics like blink rate and EAR.
- **Audio Integration for Context:** By analyzing microphone array data—vocal tone, prosody, and linguistic cues—the system captures nuances such as stress or hesitation that visuals alone may miss.
- **Unified Behavioral Profile:** The "Integrated Analysis" dashboard fuses all data streams, with a central agentic layer dynamically managing modules to deliver a real-time, coherent behavioral assessment.



References

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Thank You